

Jessica U. Ezemba

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scholar.google.com/citations?user=RNnvpBwAAAAJ | orcid.org/0009-0003-4003-1522

EDUCATION

Carnegie Mellon University Ph.D., Mechanical Engineering GPA: 3.92/4.0	Pittsburgh, PA Expected May 2027
Carnegie Mellon University M.S., Integrated Innovation for Products & Services GPA: 3.96/4.0	Pittsburgh, PA Dec. 2022
Texas A&M University B.S., Mechanical Engineering GPA: 3.52/4.0	College Station, TX May 2021

RESEARCH INTERESTS

I research at the intersection of artificial intelligence and engineering design. My work spans **vision-language models** for engineering simulations, **multi-agent orchestration systems**, **graph representation and learning**, and **neural surrogate modeling**. My future research aims to develop autonomous AI-driven design systems that integrate physical reasoning with foundation models to accelerate discovery across engineering domains. I am currently seeking summer 2027 opportunities.

PUBLICATIONS

In Review

1. **Ezemba, J.**, McComb, C., & Tucker, C. (2026). "From Organizational Theory to Agent Teams: Implementing and Evaluating MAS Architectures for Engineering Design." IDETC 2026.
2. **Ezemba, J.**, et al. (2026). "Organizationally-Intelligent Agents: Connecting Multi-Agent Engineering Systems to Organizational Design Theory." Design Computing and Cognition.
3. Mehta, M.D.*, **Ezemba, J.***, et al. (2026). "VizArQA: A Benchmark for Visual Question Answering in Architectural Simulation." J. of Eng. for Sustainable Buildings and Cities.
4. **Ezemba, J.**, Pohl, J., Tucker, C., & McComb, C. (2026). "OpenSeeSimE: A Large-Scale Benchmark to Assess Vision-Language Model QA Capabilities in Engineering Simulations." Nature Communications Engineering.

Published

5. **Ezemba, J.**, McComb, C., & Tucker, C. (2026). "Neural Network Surrogate Modeling for SFEM Using 3D Graph Representations: A Comparative Study." Journal of Mechanical Design, Vol. 148, No. 1.
6. **Ezemba, J.**, McComb, C., & Tucker, C. (2025). "Neural Network Surrogate Modeling for Stochastic FEM Using 3D Graph Representations: A Comparative Study." IDETC 2025.
7. **Ezemba, J.**, McComb, C., & Tucker, C. (2025). "Simulation vs. Hallucination: Assessing VLM QA Capabilities in Engineering Simulations." DESTION 2025, pp. 1-9.
8. **Ezemba, J.**, Afful, J., & Wang, M. (2025). "PhySiViT: A Physics Simulation Vision Transformer." SC25 Poster.
9. Tarutira, T.T., Alagbe, I., **Ezemba, J.**, Berhanu, E., & Ndashimye, E. (2024). "Baboon Detection for Smart Agriculture Using Deep Learning Algorithms." IEEE UV 2024.
10. **Ezemba, J.**, Cunningham, J.D., & Tucker, C.S. (2023). "Parameter Extraction From Images Using Multilabel Supervised Learning." IDETC 2023.
11. **Ezemba, J.**, & Layton, A. (2022). "Bio-Inspired Avenues for Advancing Brain Injury Prevention." Journal of Mechanical Design, Vol. 144, No. 12.
12. Khaled, M., Srinivasan, S., Toure, A., Chen, M., Kincaid, E., Lopaz, T., ..., **Ezemba, J.**, et al. (2021). "DREAMS: Drilling and Extraction Automated System." arXiv:2106.05874.

SKILLS

Programming & Engineering: Python, FEniCSx, PyAnsys, ANSYS, MATLAB, SOLIDWORKS, CATIA V5, AutoCAD, PTC Creo, LabVIEW

AI/ML: PyTorch, PyTorch Geometric, Multi-Agent Systems, LangGraph, Vision Transformers, GNNs, Geometric Deep Learning, Autoencoders, LLMs, RAG, MCP

Computation & DevOps: HPC, FEA, CFD, Probabilistic Simulation, Surrogate Modeling, Neural Operators, Linux, AWS

Engineering: Controller Design, Optimization Design, Product Design, Prototyping, GD&T

PHD RESEARCH

Carnegie Mellon University

Pittsburgh, PA

PhD Candidate, Mechanical Engineering

Aug. 2023 - Present

Advisors: Prof. Christopher McComb, Prof. Conrad Tucker

- Built custom Multi-Agent System (MAS) for engineering design and simulation utilizing organizational design principles, increasing computational efficiency by 50%
- Developed Model Context Protocol (MCP) server for prompt-to-CAD, parametric geometry variation, and FEA/CFD simulations, enabling faster integration into MAS and reducing computational costs by 50%
- Created comprehensive benchmark datasets (200,000+ questions, 10,000 simulations) for evaluating LLM performance on engineering simulation reasoning across text, image, and video modalities
- Engineered neural network surrogate models with FEniCSx to quantify uncertainty in complex engineering simulations, reducing computational costs by 90% across 16,000+ 3D geometries
- Advanced the application of Geometric Deep Learning to Graph Neural Networks, creating custom data loaders that effectively process complex 3D unstructured mesh geometries
- Integrated state-of-the-art LLMs and Vision Transformers for multimodal engineering data interpretation
- Architected custom self-retrieval and agentic RAG systems using LangGraph, automating simulation interpretation and reducing analysis time from days to hours

PROFESSIONAL EXPERIENCE

KLA

Ann Arbor, MI

Advanced Mechatronics Intern

May 2022 - Aug. 2022 & May 2023 - Aug. 2023

- Developed a 3D parametric geometry and physics-based optimizer to reduce the time to design flexures
- Created a 3D design exploration algorithm using ANSYS for fatigue, buckling, and frequency analysis
- Developed an optimization algorithm in MATLAB to decrease controller tuning time by over 50%
- Analyzed frequency response functions with loop shaping techniques to improve reliability of system
- Generated technical documents for the Frequency Response Analysis on MATLAB to train employees

Symbolic

Wilmington, MA

Systems Design Intern

Jan. 2023 - May 2023

- Increased warehouse automation throughput using FEA to optimize component mass and structural integrity of multiple 3D component parts
- Designed high-risk safety devices for warehouse staff, achieving 100% compliance with safety regulations
- Developed integrated documentation covering multiple subsystem aspects, improving overall system understanding

Spirit AeroSystems

Wichita, KS

Research and Development Engineering Intern

May 2021 - Aug. 2021

- Designed, prototyped, and introduced tooling aids to save over \$100,000 in safety costs in Boeing programs
- Improved current method of skin defect detection on aircraft fuselage to reduce cost of rework
- Modeled engineering drawings of aircraft tooling using CATIA V5 to better document available models

ADDITIONAL RESEARCH EXPERIENCE

Carnegie Mellon University

Pittsburgh, PA

Master Research Assistant

Jan. 2022 - Aug. 2023

- Developed multilabel CNN model to extract over 2,000 design parameters from bicycle images, achieving 96% accuracy to enable automated reverse engineering of legacy designs
- Fine-tuned pre-trained ResNet50 model on grayscale engineering drawings to predict multi-type design parameters, reducing training time and computational costs compared to training from scratch

Texas A&M University

College Station, TX

NASA Langley Research Design Project - Mechanical Design Lead

June 2020 - May 2021

- Managed 6 engineers to build prototypes to meet NASA requirements and specifications
- Designed a drill rig base to meet specific constraints and criteria while on Mars with a \$9,000 budget
- Selected as 1 of 10 universities to compete nationally in NASA Langley Research Center

Texas A&M University

College Station, TX

Biomechanical Environments Laboratory - Undergraduate Research Assistant

Aug. 2019 - May 2021

- Created CAD drawings for a pressure test chamber to help decrease death rate due to concussions
- Manufactured and assembled 7 prototype components in SolidWorks for pressure chamber
- Analyzed pressure data from 100 force impact tests to predict damage

Texas A&M University

College Station, TX

Bio-inspired Sustainable Systems Laboratory - Undergraduate Research Assistant

Oct. 2020 - May 2021

- Constructed framework for analyzing common protective mechanisms in nature from 8 organisms
- Published identified gaps and areas of improvement for protection using bioinspired methodologies

Art et Metiers

Aix-en-Provence, France

Technical Partner

May 2019 - June 2019

- Modeled an industrial mixer system using Finite Element Analysis Software, CATIA, to predict failure
- Identified failure in 3 shafts in an industrial mixer system to help reduce risk of injury

CONFERENCES / PRESENTATIONS

1. **Ezemba, J., Afful, J., & Wang, M.** "PhySiViT: A Physics Simulation Vision Transformer." SC25, St. Louis, MO, 18 Nov. 2025. Poster Presentation.
2. **Ezemba, J., McComb, C., & Tucker, C.** "Simulation vs. Hallucination: Assessing VLM QA Capabilities in Engineering Simulations." DESTION 2025, Irving, CA, 6 May 2025. NSF Sponsored Poster.
3. **Ezemba, J., McComb, C., & Tucker, C.** "Neural Network Surrogate Modeling for Stochastic FEM using 3D Graph Representations." IDETC 2025, Anaheim, CA, 18 Aug. 2025. Research Presentation.
4. **Ezemba, J., McComb, C., & Tucker, C.** "Parameter Extraction from Images Using Multilabel

Supervised Learning." IDETC 2023, Boston, MA, 18 Aug. 2023. Research Presentation.

5. **Ezemba, J.** "Traumatic Brain Injury at the Cellular Level: Design of a Concussion Bioreactor." WE20 Student Research Competition, Virtual, 6 Nov. 2020. Conference Poster.
6. **Ezemba, J.** "Traumatic Brain Injury at the Cellular Level: Design of a Concussion Bioreactor." Gulf Coast Undergraduate Research Symposium, Virtual, 31 Oct. 2020. Research Poster.

TEACHING EXPERIENCE

Carnegie Mellon University Pittsburgh, PA
TA - *Special Topics: Designing and Deploying AI/ML Systems (Graduate)* 2024

- Led full class sessions, held office hours, and created course content covering ethical AI/ML, automated machine learning, human-AI co-design, and agentic AI frameworks

Carnegie Mellon University Pittsburgh, PA
TA - *Mechanics II: 3D Design (Undergraduate, ~150 students)* 2024

- Led TA recitation sessions and office hours covering 3D statics, torsion, multiaxial stresses, failure criteria, and machine design

Texas A&M University College Station, TX
Lab Instructor - *Experimental Physics and Engineering Lab II* July 2018 - Dec. 2020

- Instructed a 3-hour mechanical physics weekly laboratory for 48 engineering students across 4 semesters

MENTORING

- Co-advised one Integrated Innovation Institute masters student and one undergraduate MechE student with PI; each mentorship led to one journal publication
- Mentored 3 CMU Africa students; two won exchange program to CMU Pittsburgh; one secured position with EU tech company; acted as PI for Baboon Detection paper
- Mentored CMU ETIM masters student who secured internship and full-time offer at Phoenix manufacturing company

AWARDS AND HONORS

NSF ACCESS PI Grant	2025
NSF Student Travel Award - DESTION 2025	2025
Best Poster Award - CAARMS	2024
DED BPart Committee Fellowship + Travel Scholarship - IDETC 2023	2023
Graduate Education for Minorities (GEM) Fellowship	2022
National Academy of Engineering (NAE) Grand Challenge Scholars	2021
Society of Women Engineers (SWE) Scholarship	2020
Benjamin Gilman Scholars Recipient	2019

SERVICE AND OUTREACH

- Human-Centered AI for K-12 Education (Moonshot Grant) - Assisted bringing 20 high school seniors from 4 PA districts to redesign K-12 education using human-centered AI with CMU faculty mentorship
- NAE Design Science & AI Workshop - Part of organizing team for official NAE-sponsored 3-day workshop in Kigali, Rwanda (2024)
- K-12 Outreach - Taught chemistry to high school students at Obama High School as part of CMU outreach
- MEGSO Mentorship Program Lead - Organized 3 industry speaker events and company visits for graduate students
- Conference Volunteering - CDFAM 2023 (NYC), IEEE Intl. Automated Vehicle Validation Conference 2024 (CMU)